Exp 10 MATLAB CODE

```
clc;
clear;
close all;

Fs = 100000;
t = 0:1/Fs:0.02;

Am = 1;
fm = 1000;
fc = 10000;
kf = 5000;
SNR = 10;

% Message Signal
m = Am*cos(2*pi*fm*t);

% FM Signal Generation
int_m = cumsum(m)/Fs;
s = cos(2*pi*fc*t + 2*pi*kf*int_m);

% Add Gaussian White Noise
noisy_signal = awgn(s,SNR,'measured');

% FM Demodulation
analytic_signal = hilbert(noisy_signal);
inst_phase = unwrap(angle(analytic_signal));
inst_freq = [diff(inst_phase)*Fs/(2*pi), 0];
```

```
% Recover Message Signal
recovered = (inst_freq - fc)/kf;
[b,a] = butter(5,2000/(Fs/2),'low');
recovered = filtfilt(b,a,recovered);

% Normalize Recovered Signal
recovered = recovered - mean(recovered);
recovered = recovered/max(abs(recovered));

% Calculate MSE
MSE = mean((m-recovered).^2);

% Plot Signals
figure;

subplot(4,1,1);
plot(t,m);
title('Original Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

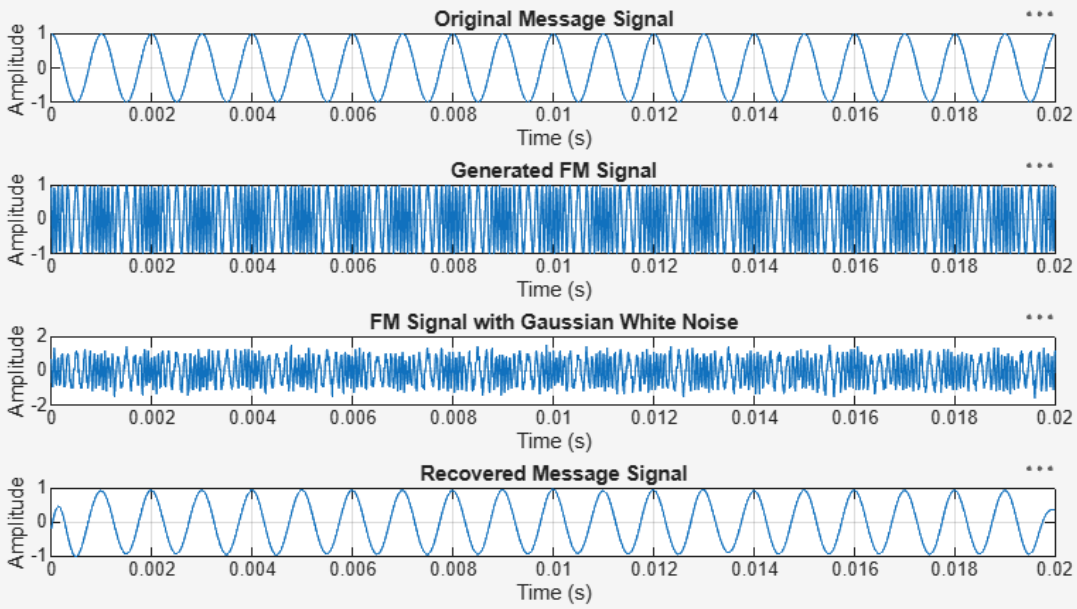
subplot(4,1,2);
plot(t,s);
title('Generated FM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;

subplot(4,1,3);
plot(t,noisy_signal);
title('FM Signal with Gaussian White Noise');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,4);
plot(t,recovered);
title('Recovered Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Performance Analysis
fprintf('Signal-to-Noise Ratio = %d dB\n',SNR);
fprintf('Message Frequency = %d Hz\n',fm);
fprintf('Carrier Frequency = %d Hz\n',fc);
fprintf('Frequency Deviation = %.2f Hz\n',kf*Am);
fprintf('FM Modulation Index = %.2f\n',kf*Am/fm);
fprintf('Mean Squared Error = %.6f\n',MSE);
```


Original Message Signal
Amplitude
Time (s)
Generated FM Signal
Amplitude
Time (s)
FM Signal with Gaussian White Noise
Amplitude
Time (s)
Recovered Message Signal
Amplitude
Time (s)